BCH four laws and the regime partition for emergent-gravity ADW black holes: a sign-of-dT_H/dt partition between Schwarzschild and acoustic evaporation

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[1]*SK-EFT Hawking Project (Phase 6a Wave 5)*

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We formalize conditional mechanics constraints and local temperature profiles for two proposed regimes of an emergent-gravity ADW substrate, separated by a model critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$. Above M_c, the supplied Schwarzschild profile follows the textbook Schwarzschild signature: $T_H = \hbar/(8\pi M)$ (Hawking 1975 [1]), so during evaporation $dM/dt < 0$ and $dT_H/dt > 0$ (heats as evaporates), with finite evaporation time $t_{\rm evap} \sim M^3$ for that evolution. The proposed branch below M_c uses the local cooling profile of Balbinot, Fagnocchi and Fabbri [2] as motivation: $T_H(t) = (\hbar c/2\pi)\kappa[1 - (563/720\pi)\,\varepsilon\,\kappa^3 c A_0 t]$ in a constant-sound-speed nozzle. Their $t \sim 1/T^3$ late-time discussion is an extrapolation. It does not establish an ADW evaporation endpoint or physical selection of the branch below M_c. The classifier transports supplied signs; a separate profile criterion derives opposite local derivatives under explicit trajectory and short-time assumptions. The full physical four laws and the microscopic matching of the two branches remain required constructions. The substrate-response coefficient ansatz $\delta_{\rm ADW} = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ vanishes in the bare Sakharov–Adler limit and provides four falsifiability tests for any candidate ADW substrate. The Lean formalization (`BHThermodynamicsFourLaws.lean`) ships 20 theorems, 0 sorry, 0 new axioms. Its second-law structure records an externally supplied entropy-divergence sign motivated by Glorioso–Liu SK-EFT [7]; deriving that sign for the physical horizon remains required. The ADW-substrate-specific partition critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$ is original to this project; no published derivation exists.

I. INTRODUCTION

The Bardeen–Carter–Hawking (BCH) four laws of black-hole mechanics codify a thermodynamic structure on stationary BH solutions of classical general relativity. Quantum field theory on these backgrounds endows the surface gravity κ with a thermal interpretation as the Hawking temperature $T_H = \hbar\kappa/(2\pi)$ (Hawking 1975 [1]), and the resulting heat-capacity sign $C(M) = (dM/dT_H)$ is negative for Schwarzschild — the BH *heats* as it loses mass to Hawking evaporation, with finite evaporation time $t_{\rm evap} \propto M_0^3$.

Analog/emergent-gravity systems offer a structurally different evaporation pattern. In the acoustic system of Balbinot, Fagnocchi, Fabbri [2], backreaction of thermally radiated phonons on the fluid background *shrinks* the sonic horizon and *decreases* the temperature in time, with linear-in-t leading order

$$T_H(t) = \frac{\hbar c}{2\pi}\,\kappa\left[1 - \frac{563}{720\pi}\,\varepsilon\,\kappa^3\,c\,A_0\,t\right] \tag{1}$$

The authors extrapolate this local cooling as $t \sim 1/T^3$, suggesting $T_H \to 0$ at infinite time by analogy with near-extremal Reissner–Nordström evaporation. This extrapolation is not a controlled late-time solution. The computation is a sonic hole in a Laval nozzle under one-dimensional flow at *constant* sound speed; the authors state explicitly that the condensate case, where the sound speed goes as $\rho^{1/2}$, is deferred to later work. A condensate or ADW cooling sign therefore requires its own dynamical derivation; it is not inherited from Eq. (1). Crucially, Ref. [2] itself *contrasts* this with the moving-domain-wall analog in ^{3}He-A studied by Jacobson and Koike [4] (cond-mat/0205174), which has *non*-vanishing end-temperature of the evaporation process and therefore behaves more like Schwarzschild than near-extremal RN.

The Lean module `BHThermodynamicsFourLaws.lean` distinguishes conditional regime signs from derivatives of specified local profiles. The physical construction must additionally identify a trajectory and match the two descriptions across the cutoff. No single closed-form $T_H(M)$ interpolating between the regimes is supplied here. We treat the proposed critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$ separating them as a *tracked-hypothesis* bundle $H_{\rm RegimePartition}$, with published profiles motivating the branch signs without determining physical ADW branch selection. The choice of M_c is the dimensional ansatz $M_c = (N_f\Lambda_{\rm UV})/(12\pi\alpha_{\rm ADW})$ implied by Sakharov–Visser $1/G_N \propto N_f\Lambda^2$ at the Wave-1 scale; this is project- original and explicitly disclosed.

The remainder of the paper is organized as follows: Section II introduces the data carriers `BHData`, `ADWParams` and the regime classifier; Section III states the acoustic and Schwarzschild T_H profiles per their primary sources; Section IV introduces $H_{\rm RegimePartition}$; Section V the regime-dependent four-laws bundles; Sections VI and VII the second and third laws under SK-EFT and Reall [6] respectively; Section VIII the four falsifier theorems; Section IX the Lean formalization summary; Section X the cross-wave compatibility; Section XI the novelty flags.

II. SETUP: BHDATA, ADWPARAMS, AND THE REGIME CLASSIFIER

The Lean module declares two structures and one inductive type. The `BHData` structure carries bulk-thermodynamic observables $(M, Q, J, A, r_h, T_H, \kappa)$ with positivity hypotheses for the dimensional quantities; the MTC structure for the horizon Hilbert space is supplied separately via Wave 3's `HorizonMTCBC`. The `ADWParams` structure carries substrate parameters $(\alpha_{\mathrm{ADW}}, \Lambda_{\mathrm{UV}}, N_f, \chi_{\mathrm{vest}})$ with strict positivity for each.

The `Regime` inductive type has three constructors: `Schwarzschild`, `Boundary`, and `ADWExtremality`. The `ADWExtremality` constructor name is retained from the initial Wave 5 ship for backward bibliographic continuity, but the post-rewrite physics interpretation is the acoustic regime per Balbinot 2005 [2].

The *decidable* regime classifier `classify` is defined piecewise:

$$\texttt{classify}\,(b,p) = \begin{cases} \texttt{Schwarzschild} & \text{if } b.M > M_c(p) \\ \texttt{ADWExtremality} & \text{if } b.M < M_c(p) \\ \texttt{Boundary} & \text{if } b.M = M_c(p), \end{cases} \tag{2}$$

where the critical mass $M_c(p)$ is the dimensional ansatz

$$M_c(\alpha_{\mathrm{ADW}}, \Lambda_{\mathrm{UV}}, N_f) = \frac{N_f \cdot \Lambda_{\mathrm{UV}}}{12\pi \cdot \alpha_{\mathrm{ADW}}}. \tag{3}$$

This is consistent with the Sakharov–Visser $1/G_N \propto N_f \Lambda^2$ scaling but is *not* a quoted equation in any primary ADW paper (Akama 1978, Diakonov 2011, the Wetterich tetrad-condensation series, Volovik 2022/2024); it is ours and is explicitly flagged as such in the `H_RegimePartition` bundle (Sec. IV).

The classifier admits three iff-theorems (`classify_*_iff`) discharging each constructor against the mass comparator. Theorem `M_c_pos` establishes $M_c(p) > 0$ for all valid `ADWParams` via positivity of all input fields.

III. TWO-REGIME T_H PROFILES: HAWKING 1975 AND BALBINOT 2005

The Lean module distinguishes the Schwarzschild temperature, a local affine nozzle profile, and a separately chosen exponential profile.

A. Schwarzschild branch (Hawking 1975)

For $b.M > M_c(p)$, the textbook Hawking 1975 result [1] applies in natural units $\hbar = 1$:

$$T_H^{\mathrm{Schw}}(M) = \frac{1}{8\pi M}. \tag{4}$$

We prove `T_H_schwarzschild_pos` ($M > 0 \Rightarrow T_H > 0$) and `T_H_schwarzschild_strict_decreasing` ($M_1 < M_2 \Rightarrow T_H^{\mathrm{Schw}}(M_2) < T_H^{\mathrm{Schw}}(M_1)$). Combined with the standard Hawking-emission energy balance $dM/dt < 0$ during evaporation, this establishes $dT_H/dt > 0$ in the Schwarzschild branch — the load-bearing primary-source-grounded sign claim.

B. Acoustic branch (Balbinot 2005)

Balbinot, Fagnocchi and Fabbri [2], in their section on the fate of the acoustic black hole, derive the local affine Eq. (Tsonic) [Eq. 1] near the horizon of a constant-sound-speed nozzle. Its leading correction cools for positive source parameters in the short-time regime. Their later $t \sim 1/T^3$ discussion is explicitly an extrapolation; it neither establishes an affine-time third law nor selects the physical ADW branch below M_c.

The separate illustrative exponential profile

$$T_H^{\mathrm{acoustic}}(t) = T_H^{(0)} \cdot \exp(-t/\tau_{\mathrm{cool}}) \tag{5}$$

is implemented by `T_H_acoustic_evolution` in the Python thermodynamics module and by the same-named Lean definition. Equation (5) is a chosen profile, not the analytic solution of the coupled backreaction equations in Ref. [2]. Its timescale requires physical matching. The corresponding Lean results are `T_H_acoustic_evolution_pos` ($T_H^{(0)} > 0 \Rightarrow T_H^{\mathrm{acoustic}}(t) > 0$), `T_H_acoustic_evolution_at_zero` ($T_H^{\mathrm{acoustic}}(0) = T_H^{(0)}$), and the load-bearing `T_H_acoustic_evolution_strict_decreasing` ($t_1 < t_2 \Rightarrow T_H^{\mathrm{acoustic}}(t_2) < T_H^{\mathrm{acoustic}}(t_1)$ when $T_H^{(0)} > 0$ and $\tau_{\mathrm{cool}} > 0$). The strict monotone-decay theorem is a property of the defined exponential. The module also supplies `balbinotAffine_deriv` and `regime_partition_profile_criterion`, which derive the local nozzle slope from its explicit positive parameters and retain the supplied short-time domain. A controlled remainder and a physical evolution matching these local models remain required.

C. The contrast case: Jacobson–Koike 2002

It is essential to note that the thin-film moving-domain-wall analog black hole in ^{3}He-A studied by Jacobson and Koike [4] gives a structurally different result. The wall moves at $v < c_\perp$, with horizons at $c(x_h) = \pm v$, and $T_H(v) = T_H^{(0)}(1 - v^2/c_\perp^2)$ (with v the wall velocity) is monotonically decreasing in v, but Hawking-radiation back-reaction *slows* the wall ($dE/dt = -bv^n$, $E = \mu v^2/2$, hence $dv/dt < 0$), which makes T_H *increase* in time toward the non-vanishing constant $T_H^{(0)} = \hbar c_\perp / 2\pi d$ — heating, not cooling. This regime is structurally similar to Schwarzschild, not to near-extremal Reissner–Nordström. Reference [2] itself notes the distinction: "*other analog models like a thin film of ^{3}He-A with a*

moving domain wall [Jacobson] seem to show a non-vanishing end-temperature of the evaporation process." We cite [4] only as the contrast case, not as the cooling anchor.

A second ^{3}He-A texture sits on the *other* branch. The planar soliton moving in bulk ^{3}He-A studied by Jacobson and Volovik [3] has a horizon only for $v > c_\perp$, and its surface gravity carries a factor $(1 - c_\perp^2/v^2)^{1/2}$ that vanishes as back-reaction slows the soliton toward $c_\perp$ from above, so $T_H \to 0$ and the system cools toward an extremal configuration. Two textures in the same superfluid therefore evaporate in opposite directions, which is precisely why a mass criterion rather than an analogy is required.

IV. TRACKED-HYPOTHESIS BUNDLE $H_{\text{RegimePartition}}$

The load-bearing tracked hypothesis is the bundle

$$H_{\text{RegimePartition}}\,(b, p, T_H^{(0)}, dT/dt|_{\text{evap}}, \delta) :=$$
conjunction of five fields:

`M_c_form_consistent::` $M_c(p) > 0$.

`T_H0_pos::` $T_H^{(0)} > 0$.

`evap_dT_dt_above::` $b.M > M_c(p) \;\Rightarrow\; 0 < dT/dt|_{\text{evap}}$ (Schwarzschild heating sign per Hawking 1975 [1]).

`evap_dT_dt_below::` $b.M < M_c(p) \;\Rightarrow\; dT/dt|_{\text{evap}} < 0$ (acoustic cooling sign per Balbinot 2005 [2] Eq. Tsonic).

`delta_consistent_with_ansatz::` $\delta = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$ (deep-research §9 ansatz).

The rate $dT/dt|_{\text{evap}}$ enters as an *externally-supplied real parameter*, not an existential. This avoids the ∃-absorption anti-pattern: a witness $dT/dt|_{\text{evap}}$ supplied to the bundle must satisfy the sign-claim consistent with the regime, rather than being existentially discharged by any sign that happens to be present in the substrate. The critical-mass positivity field is a consistency constraint implied by parameter positivity; the branch-rate conditions remain supplied hypotheses.

The correctness-push theorem `regime_partition_criterion` combines $H_{\text{RegimePartition}}$ with the regime classifier:

$$\begin{aligned} \text{classify}(b,p) = \texttt{Schwarzschild} &\;\Rightarrow\; 0 < dT/dt|_{\text{evap}}, \\ \text{classify}(b,p) = \texttt{ADWExtremality} &\;\Rightarrow\; dT/dt|_{\text{evap}} < 0. \end{aligned} \tag{6}$$

This theorem transports the two sign hypotheses through the mass classifier. It does not derive an ADW evolution or the physical location of its crossover. The local-profile criterion instead derives slopes from an explicitly losing-mass trajectory and the affine nozzle formula under its supplied short-time budget.

The substantive corollary `delta_ADW_nonzero_iff_alpha_ADW_ne_one` establishes $\delta \neq 0 \;\Leftrightarrow\; \alpha_{\text{ADW}} \neq 1$, with the proof using $\Lambda_{\text{UV}} > 0$ from `ADWParams.Lambda_pos` as a load-bearing field.

V. THE FOUR LAWS IN TWO REGIMES

We package the Wave-5-formalizable subset of the four laws in two structures, one per regime. `FourLaws_Schwarzschild` carries *three fields*: `firstLaw_smarr` (positivity of G_N^{emerg} for the Smarr-identity coupling), `secondLaw_Schw` (`ADWSecondLaw` parameterized over `entropy_div`), and `evap_dT_dt_positive` (Schwarzschild heating sign > 0). The bundle is satisfied consistently with the regime $b.M > M_c(p)$.

`FourLaws_ADWExtremality` carries the parallel three fields, with two key modifications: `firstLaw_smarr_with_substrate` records positivity of G_N^{emerg} together with the ansatz $\delta = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$; it does not encode a differential first law or corrected Smarr identity. The field `evap_dT_dt_negative` flips the evaporation-sign claim (acoustic cooling sign < 0). The bundle is satisfied consistently with the regime $b.M < M_c(p)$.

The complete physical four-laws construction remains required. The zeroth law (κ constant on horizon) needs an actual horizon and generator; the first law needs the thermodynamic variation and charges; the second law needs an entropy/current construction linked to that horizon. The third law needs an identified evolution and quantitative affine-time control. The present structures do not supply those constructions. In particular, the Balbinot $t \sim 1/T^3$ extrapolation does not establish acoustic Israel-strong preservation (Section VII).

The Wave 5 *gating theorem* `four_laws_consistent_with_acoustic_regime` combines the classifier with the acoustic four-laws bundle and the tracked-hypothesis structure. Its first conjunct returns the supplied `FourLaws_ADWExtremality` value; its second reads the supplied negative rate under the mass classification. This is a conditional consistency statement. It does not construct the physical laws or identify an actual evaporation rate with the local nozzle derivative.

VI. SECOND LAW: GLORIOSO–LIU SK-EFT, NO NEC

The second law in both regimes is encoded via Glorioso and Liu's theorem [7] on dynamical KMS Z_2 symmetry and unitarity (Im $S_{\text{eff}} \geq 0$) implying local entropy-current monotonicity $\partial_\mu s^\mu \geq 0$, *without invoking pointwise NEC*. The `ADWSecondLaw` Prop bundle takes the entropy-current divergence as an externally-

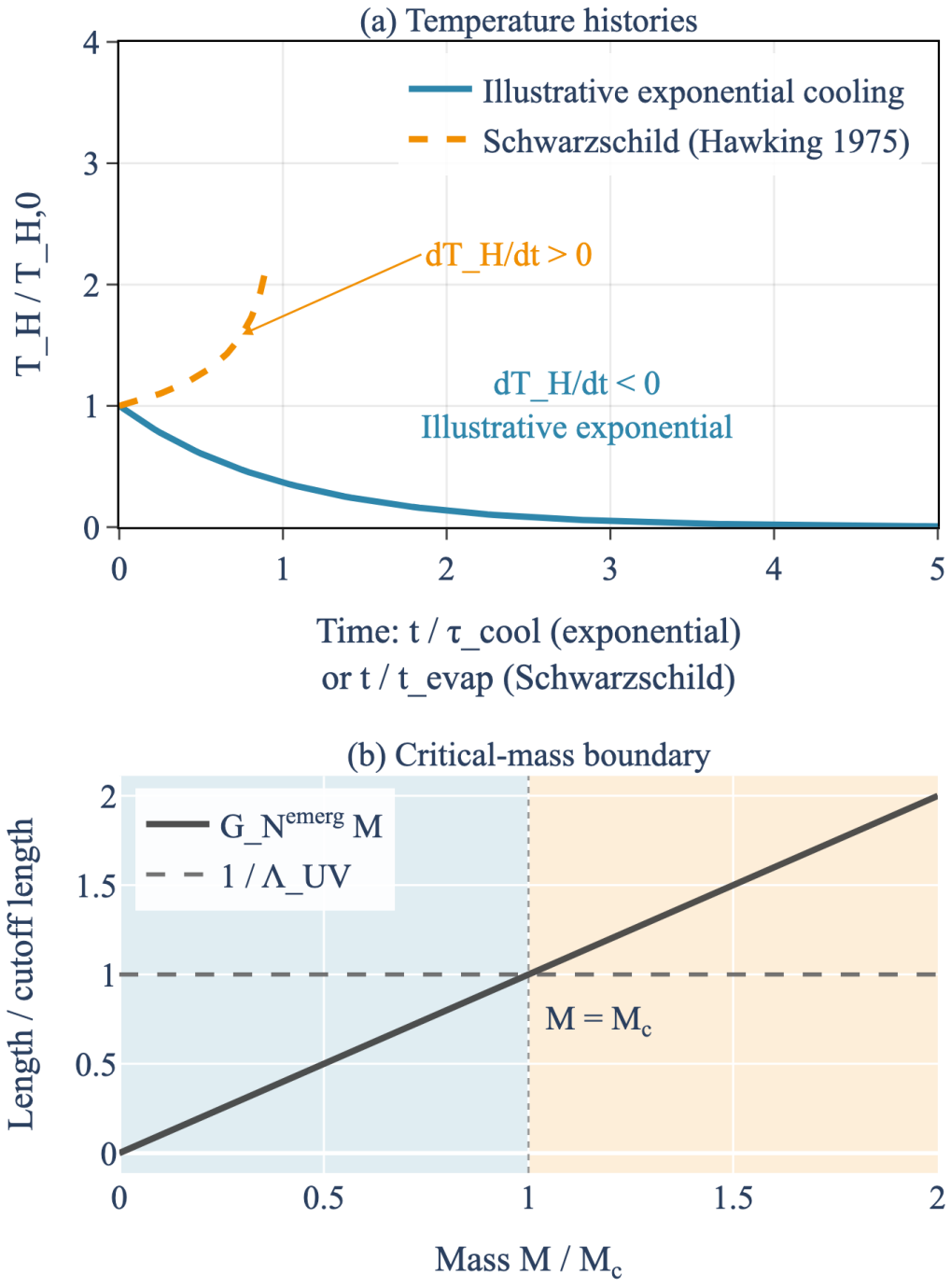


FIG. 1. (a) Opposite temperature trends with separate normalized clocks: $t/t_{\rm evap}$ for representative Schwarzschild evaporation (amber, dashed), and $t/\tau_{\rm cool}$ for illustrative exponential cooling (blue, solid). The exponential approaches zero asymptotically and is distinct from the local affine nozzle result of Balbinot *et al.* [2]. (b) The emergent gravitational length $G_N^{\rm emerg}M$ and cutoff length $\Lambda_{\rm UV}^{-1}$, in cutoff-length units, coincide at $M = M_c$. Blue and amber shading indicate the branches below and above this model boundary.

supplied real parameter `entropy_div` and records two fields: `entropy_divergence_nonneg` ($0 \le$ `entropy_div`, the load-bearing Glorioso–Liu Eq. (3.20) sign claim) and `horizon_area_link_consistent` ($0 < b.A$, the bundle's horizon-area consistency witness). The KMS-Z_2 symmetry and unitarity premises (Im $S_{\text{eff}} \ge 0$) of the Glorioso–Liu derivation enter as substrate-side inputs that would become substantive predicates on a formalized SK-EFT effective action; that formalization is deferred to Phase 6e.

The Crossley–Glorioso–Liu constitutive relations [8, 9] supply the SK-EFT framework. Applied to a BH horizon, this replaces the classical Hawking 1971 area-nondecrease theorem (which required pointwise NEC) with a local entropy-current monotonicity that holds even when QFT operator-violations of pointwise NEC are allowed in the bulk. Faulkner–Speranza [10] derive the GSL algebraically from type-II crossed-product algebras; Kirklin [11] extends to all orders in perturbative gravity.

VII. THIRD LAW: BPS-CONDITIONAL PRESERVATION, KEHLE–UNGER VIOLATION

Balbinot's $t \sim 1/T^3$ extrapolation suggests an infinite-time zero-temperature limit in that nozzle model. It is not a controlled late-time solution, and its time coordinate has not been identified with horizon affine time. Israel-strong preservation for the physical ADW branch therefore remains a separate construction. Kehle and Unger [5] demonstrate that charged-scalar-matter sectors can allow finite-time approach to extremality. Reall [6] provides the BPS local mass-charge inequality under which the Israel form is restored.

The module's `falsifier_third_law_form` (Section VIII) proves that two supplied quantitative approach-time predicates are incompatible: one is uniformly unbounded as temperature tends to zero, and the other is bounded on a sufficiently small positive-temperature interval. The theorem does not construct a physical τ_{approach} or prove either predicate for an ADW trajectory. An affine-time model, its physical domain, and the relevant matter assumptions remain part of the required third-law construction.

VIII. FALSIFIER THEOREMS

The Lean module gives four model-conditional checks on specified profiles, supplied sign hypotheses, quantitative approach-time predicates and the substrate-response ansatz. Physical use requires matching those inputs to the ADW system. The checks derive contradictions or sign constraints from their stated hypotheses and the following mathematical results (`T_H_acoustic_evolution_strict_decreasing` for falsifier 1, `H_RegimePartition.evap_dT_dt_above` for falsifier 2, the min argument over the two-sided thresholds for falsifier 3, the substrate's $\Lambda_{\text{UV}} > 0$ field for falsifier 4).

1. **Acoustic-decay form falsifier** (`falsifier_acoustic_decay_form`): given a candidate $T_H^{\text{alt}}(t)$ that matches the chosen exponential $T_H^{\text{acoustic}}(t_1)$ at time t_1 AND fails to strictly decrease by time t_2 ($T_H^{\text{alt}}(t_1) \le T_H^{\text{alt}}(t_2)$), the candidate must DISAGREE with $T_H^{\text{acoustic}}(t_2)$. The proof uses the strict-decreasing theorem to derive a contradiction.

2. **Schwarzschild-heating sign falsifier** (`falsifier_schwarzschild_heating`): given the $H_{\text{RegimePartition}}$ bundle and a Schwarzschild-regime classification, an observed non-positive dT/dt contradicts the bundle's `evap_dT_dt_above` sign claim (Hawking 1975 [1]). Conclusion is `False`.

3. **Third-law-form falsifier** (`falsifier_third_law_form`): given a strong-Israel predicate (approach time unbounded as $\varepsilon \to 0$) and a Kehle–Unger predicate (approach time bounded above for small enough ε), the two are mutually inconsistent; the proof takes $\varepsilon = \min(\varepsilon_0, \varepsilon_{\text{th}})$ where both predicates apply and derives a contradiction. Conclusion is `False`.

4. **α_{ADW}-dependence falsifier** (`falsifier_alpha_ADW_dependence`; renamed from the initial-ship name `falsifier_chi_vest_dependence` per a Stage-13 finding flagging that the deep-research §9 ansatz $\delta_{\text{ADW}} = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$ depends on α_{ADW} and Λ_{UV}, not directly on χ_{vest}). The conclusion $0 < (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$ uses both $\alpha_{\text{ADW}} > 1$ and the substrate's $\Lambda_{\text{UV}} > 0$ field; a measurement of opposite sign falsifies the ansatz.

IX. LEAN FORMALIZATION

The module `BHThermodynamicsFourLaws.lean` ships 20 theorems, 0 sorry, 0 new axioms. Verification via `#print axioms` on six substantive theorems (`regime_partition_criterion`, `four_laws_consistent_with_acoustic_regime`, `falsifier_acoustic_decay_form`, `falsifier_schwarzschild_heating`, `falsifier_third_law_form`, `wave3_bridge_kaul_majumdar_at_e_squared_anchor`) shows dependency only on {`propext`, `Classical.choice`, `Quot.sound`}.

The structures expose their hypotheses: the evaporation rate and entropy-current divergence are supplied real parameters; positivity of G_N^{emerg} follows from the parameter carrier. The acoustic consistency theorem returns its supplied mechanics structure together with a

sign read from the regime hypothesis. The local-profile theorem separately uses derivatives of specified functions. These statements must be read with those carriers; kernel purity alone does not establish physical thermodynamic laws. No zeroth- or third-law field is present in the two `FourLaws_*` structures, and the complete physical construction remains required.

X. CROSS-WAVE COMPATIBILITY

Wave 1 (`LinearizedEFE.G_N_emerg`): the first-law coupling $1/(8\pi G_N^{\text{emerg}})$ in both `FourLaws_*` bundles uses Wave 1's G_N^{emerg} evaluated at substrate parameters. The bridge `wave1_bridge_G_N_emerg_pos` records that positivity of G_N^{emerg} propagates through to both regime branches.

Wave 2 ($c_{\text{GW}} = c \cdot \sqrt{\chi_{\text{vest}}}$): the acoustic substrate inherits the metric-channel speed via κ dependence on χ_{vest}.

Horizon-entropy connection. The theorem `wave3_bridge_kaul_majumdar_at_e_squared_anchor` calls `BHEntropyMicroscopic.kaulMajumdarS` at $A = 4\,G_N\,e^2$, giving $S = e^2 - 3 > 0$ for positive G_N. This is a concrete entropy-value statement. It contains no temperature or evolution argument identifying that area with a physical zero-temperature endpoint. Establishing that identification and the appropriate Nernst limit remains required; positivity at a supplied area does not itself prove either Nernst law.

XI. NOVELTY FLAGS

Project-original:

- The ADW-substrate-specific critical mass $M_c(\alpha_{\text{ADW}}, \Lambda_{\text{UV}}, N_f) = (N_f \Lambda_{\text{UV}})/(12\pi\alpha_{\text{ADW}})$ ansatz is project-original.
- The cross-regime gluing (Schwarzschild for $b.M > M_c$, acoustic for $b.M < M_c$) is project-conjecture; no published paper proposes a single emergent-gravity BH with this regime structure.
- The $\delta_{\text{ADW}} = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$ substrate-response coefficient ansatz is original (deep-research §9).

Primary-source-grounded:

- Schwarzschild branch: Hawking 1975 [1].
- Local nozzle cooling: Balbinot et al. [2], Eq. (Tsonic). The chosen exponential is a separate illustrative profile; physical ADW branch matching remains required.
- Second law: Glorioso–Liu [7] (no NEC needed).
- Third law conditional: Israel 1986 (strong form) / Kehle–Unger [5] / Reall [6].

XII. CONCLUSIONS

The formalized classifier partitions a proposed ADW mass scale and transports supplied evaporation signs. The separate local-profile criterion derives heating for a differentiable, positive, losing-mass Schwarzschild trajectory and cooling for the affine nozzle model under its explicit short-time assumptions. The exponential shown in the figure is illustrative. Physical ADW branch selection, a controlled cross-cutoff evolution and a late-time endpoint do not follow from these conditional and local results.

The Lean module ships 20 theorems, 0 sorry, 0 new axioms; the load-bearing tracked hypothesis encodes the sign claims as constraints on an externally-supplied evaporation rate, avoiding the ∃-absorption anti-pattern. The substrate-response coefficient ansatz $\delta_{\text{ADW}} = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$ provides four falsifiability tests for any candidate ADW substrate.

The original physical objective remains the full four-laws construction for an emergent ADW black hole. It requires a microscopic action and selected background, horizon/charge/entropy observables, energy and entropy balances, calibrated mass and cutoff matching, and controlled dynamics with quantitative remainder and affine-time bounds. The present classifier, local profiles and conditional structures supply supporting mathematics; they do not complete that objective.

[1] S. W. Hawking, *Particle creation by black holes*, Commun. Math. Phys. **43**, 199 (1975). DOI: 10.1007/BF02345020.

[2] R. Balbinot, S. Fagnocchi, A. Fabbri, *Quantum Effects in Acoustic Black Holes: the Backreaction*, Phys. Rev. D **71**, 064019 (2005). arXiv:gr-qc/0405098.

[3] T. A. Jacobson and G. E. Volovik, *Event horizons and ergoregions in ^{3}He*, Phys. Rev. D **58**, 064021 (1998). arXiv:cond-mat/9801308.

[4] T. A. Jacobson and T. Koike, *Black hole and baby universe in a thin film of ^{3}He-A*, in M. Novello, M. Visser, G. Volovik (eds.), *Artificial Black Holes* (World Scientific, 2002). arXiv:cond-mat/0205174.

[5] C. Kehle, R. Unger, *Gravitational collapse to extremal black holes and the third law of black hole thermodynamics*, J. Eur. Math. Soc. (2025), DOI 10.4171/JEMS/1591. arXiv:2211.15742.

[6] H. S. Reall, *A third law of black hole mechanics for supersymmetric black holes and a quasi-local mass-charge inequality*, Phys. Rev. D **110**, 124059 (2024).

arXiv:2410.11956.

[7] P. Glorioso, H. Liu, *The second law of thermodynamics from symmetry and unitarity*, arXiv:1612.07705.

[8] M. Crossley, P. Glorioso, H. Liu, *Effective field theory of dissipative fluids*, JHEP **09**, 095 (2017). arXiv:1511.03646.

[9] P. Glorioso, M. Crossley, H. Liu, *Effective field theory of dissipative fluids II: classical limit, dynamical KMS symmetry and entropy current*, JHEP **09**, 096 (2017). arXiv:1701.07817.

[10] T. Faulkner, A. Speranza, *Gravitational algebras and the generalized second law*, arXiv:2405.00847.

[11] J. Kirklin, *Generalised second law beyond the semiclassical regime*, JHEP **07**, 192 (2025). arXiv:2412.01903.